
An Investigation of Autorotating Probes For Venus Exploration

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Abstract

An autorotating single-winged probe design has been proposed for use in an upcoming mission to study Venus. This project supports design development by producing a numerical workflow to predict descent velocity and orientation of such probes in settled flight. This was achieved with a combination of computational fluid dynamics and rigid body dynamics, and validated with experimental tests. A large parameter sweep of different rotation speeds, orientations and terminal velocities, was conducted to evaluate forces and moments acting on a model probe. Using this workflow on a prototype probe design, a terminal velocity of $3.18 \pm 0.25 \text{ ms}^{-1}$ was found, in agreement with experimental measurements within uncertainties. Results of the parameter sweep suggest that settled flight may in fact be oscillatory about a steady equilibrium region.

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1 Introduction

The potential for life on Venus has been the subject of scientific debate for many decades [1]. Detection of phosphine in 2020, a chemical primarily produced by biological means on Earth, has renewed fervour in the debate [2]. Radio telescope data from Earth has provided valuable insight, but definitive evidence requires *in-situ* sampling. This motivates a return to Venus, last explored by the *Venera* missions in 1985, to conclusively state whether or not Venus is home to some form of microbial life.

The surface of Venus is the hottest in the Solar system. At over 700 K, it is hot enough to melt most metals, not a likely candidate for life [3]. However, the cloud layer of Venus exhibits Standard Temperature and Pressure (STP), analogous to conditions on Earth. The Morning Star Mission, led by the Massachusetts Institute of Technology (MIT), in conjunction with Rocket Lab, aims to return to Venus and sample these clouds (Figure 1) [4].

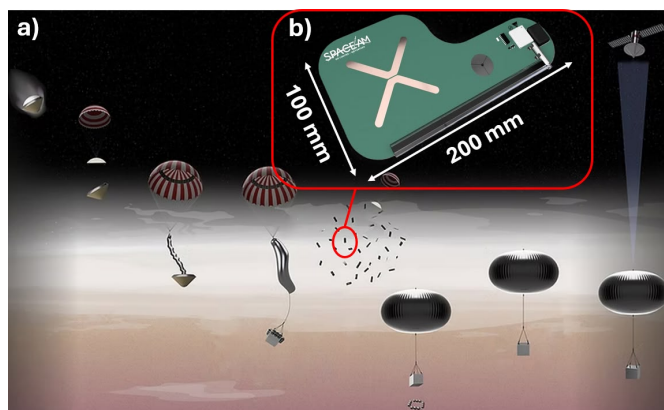


Figure 1: a) Atmospheric insertion followed by parachute, mini-probe, and balloon deployment. b) Mini-probe concept art and dimensions. Credit SpaceAM.

Collecting data on a large scale is key to the success of this mission. Scheduled for launch in 2031, the main probe will enter the atmosphere and deploy a balloon to keep scientific instruments afloat at the altitude of interest. To maximise data collection, the mission will utilise ‘mini-probes’ that will

disperse with the wind and sample a broad area of atmosphere. This study focuses on a bio-inspired design proposed by SpaceAM, which draws on the mechanics of single-winged seeds, or “samaras” [5].

Samaras are the winged fruits of trees, such as the sycamore maple in Figure 2. They use autorotation to generate lift. This reduces terminal velocity and allows the wind more time to disperse the seeds from the parent tree [6]. In samaras this prevents competition over resources. On Venus, this helps us sample data across a wider area than the balloon alone. This is achieved by increasing time-of-flight, giving the wind more time to scatter the probes. This autorotating design offers a reliable and reduced mass alternative to parachute and propeller-driven designs.

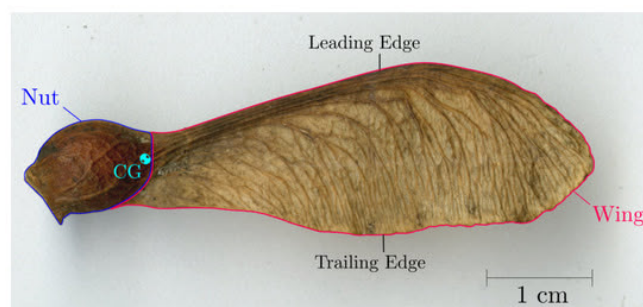


Figure 2: Sycamore seed with leading edge labelled. This is the edge that rotates forward, into the flow. Credit Jung [7].

SpaceAM approached the University of Exeter to help study the flight of these probes. Previous work by Kate Seeger (University of Exeter) led to the development of a robust workflow to run computational fluid dynamics (CFD) simulations of probes in settled rotating flight. The primary objective of this project is to take that workflow, refine it by combining aerodynamics and inertial forces, and incorporate it into a numerical pipeline capable of determining the steady-state flight profile for any given autorotating probe geometry.

This work will describe the combination of CFD simulations with rigid body dynamics, the automa-

tion of the parameter sweep to find the steady-state configuration, and the validation of the numerical pipeline via experiment and video analysis.

2 Theory

Single-winged autorotating fliers rely on the generation of a vortex, a tornado-like rotational flow, by the leading edge of the wing [8]. This vortex follows the wing, creating a region of low pressure above it that increases lift¹. The motion is self-stabilising, wing rotation generates a moment that acts to maintain the rotation rate, which contributes to lift [10]. Flow can be characterised by the ratio of inertial to viscous forces, the Reynolds number, Re :

$$Re = \frac{UL}{\nu} \quad (1)$$

where ν is the kinematic viscosity, and U and L are characteristic speeds and length scales respectively. Samaras found in nature are on the order of ~ 1 cm wide and experience flight at low Reynolds numbers, $Re \approx 100$ -260 [11]. Flow is deemed turbulent at $Re > 3000$. Turbulence is characterised by random eddies and chaotic motion, where slight changes in initial conditions lead to vastly different outcomes. This means that scaling up the model will not necessarily translate all the physics. The design proposed by SpaceAM has ten times the width of an average samara (Figure 3). Jung [12], modelled wings of up to 6.63 cm width with success, but part of our investigation will be to determine if the 10 cm width wings are feasible here.

¹There are competing theories about the mechanisms that produce lift, such as vortex “reducing stall” in Nabawy [9], but we do not investigate the mechanisms themselves here, only observable changes in flight.

2.1 Steady-State Flight

We are trying to find the steady-state flight configuration of our probes. The variables that define that configuration are: **rotation speed**, ω , **terminal velocity**², v_t , and **cone** and **pitch** angles, θ , ϕ . We use a right-handed coordinate system, so a positive moment about y causes the wingtip to tip “upwards”. θ is the angle between the span (from nut to tip) and the $x - y$ plane, and ϕ is the tilt of the chord (wing width). Negative ϕ means the leading edge is pointing down, towards the direction of motion.

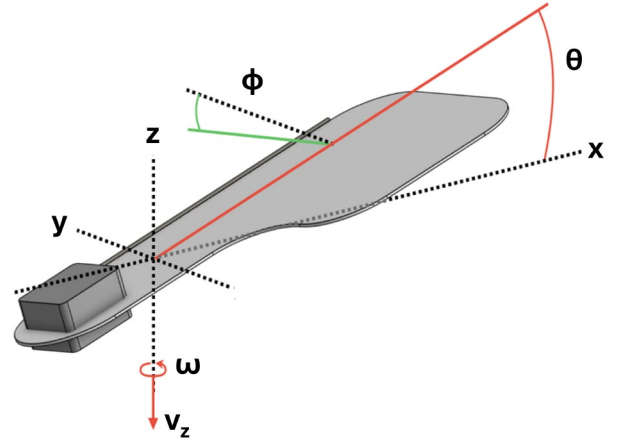


Figure 3: The probe design with annotated variables. The “box” represents where hardware for spectroscopy will be. Additional mass along the “leading-edge”, which always rotates forward into the flow. Same dimensions as in Figure 1b. Credit James O’Sullivan.

Steady-state is achieved when the net moments and forces equal zero,

$$\sum \vec{M} = 0 \quad (2)$$

$$F_z = mg \quad (3)$$

where $\vec{M}$ is the vector describing moments in x, y, z and $\vec{M} = \vec{r} \times \vec{F}$. A moment describes the tendency of an object to rotate about an axis. The summation

²In this report, v_t refers to *terminal* velocity, and v_z refers to some descent velocity that is not necessarily terminal but simply an input variable.

refers to moments that come from the aerodynamics, and from rigid body dynamics. In our approach we will simulate the probe rotating and falling at different speeds, in different orientations. None of these simulations are guaranteed to be steady-state, but the one closest to steady-state will best satisfy equations 2 and 3.

2.2 A Falling, Rotating, Rigid Object

Consider a system of two spherical masses connected by a massless rod that is both rotating and falling with air resistance in the z -direction. This simple analytical model is instructive when thinking about the forces and moments acting on our Venus probe. Centrifugal forces act perpendicular to the axis of rotation and drag forces act opposite to the direction of motion. We assume that rotation occurs about the centre of mass, the validity of this assumption is explored in Section 4.3.

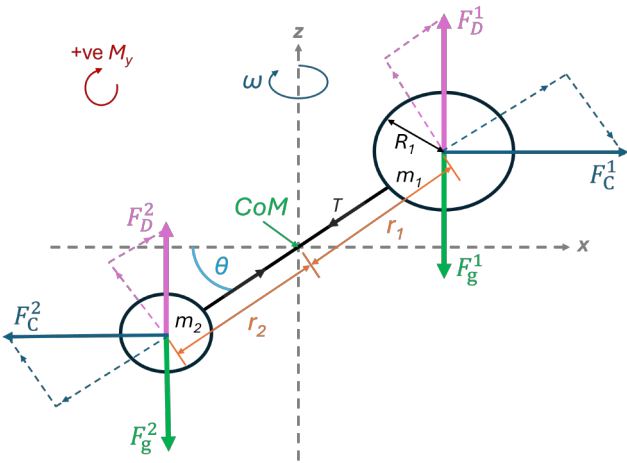


Figure 4: Schematic of the simple rotating masses model. Dashed lines indicate the components of the forces. CoM (centre of mass) lies on the axis of rotation. Clockwise moments are taken as positive.

F_g^i , F_C^i , and F_D^i represent forces due to gravity, centrifugal, and drag respectively on the i th mass, and θ is the angle between the rod and the x -axis. The system in Figure 4 will tend towards a stable solution where the body descends like a pin, vertically,

with one mass ahead of the other. If the centrifugal forces are sufficient there exists another solution such that the rod is balanced at an angle $\theta \neq 90^\circ$. Using the equations of drag and centrifugal force, we can calculate the moment about the y -axis, M_y (into the $x-z$ plane),

$$M_y = \frac{\omega^2 \sin(2\theta)}{2} (m_1 r_1^2 + m_2 r_2^2) - \frac{C_D \rho v_z^2 \cos(\theta)}{2} (A_1 r_1 - A_2 r_2) \quad (4)$$

where ω is the rotation speed, C_D is the coefficient of drag, ρ the fluid density, $A_{1,2}$ are areas facing the vertical flow, $m_{1,2}$ are masses, $r_{1,2}$ are distances from the axis of rotation, and v_z is descent velocity. This work is focused on the steady-state solution of our probe, where $M_y = 0$. For our simple model in steady-state:

$$v_z = \sqrt{\frac{2g(m_1 + m_2)}{C_D \rho (A_1 + A_2)}} \quad (5)$$

$$\omega^2 \sin(\theta) = g \frac{m_1 + m_2}{A_1 + A_2} \frac{A_1 r_1 - A_2 r_2}{m_1 r_1^2 + m_2 r_2^2} \quad (6)$$

where g is gravitational acceleration. The right-hand side of equation 6 are fixed physical parameters, so increasing ω decreases θ . This inversely proportional relationship is found in the literature on samaras, showing that with a simple model we can begin to understand real-world relationships between variables, and we can use this to test the physical consistency of our results [13].

2.3 Computational Fluid Dynamics

CFD is the use of numerical methods to analyse fluid flow problems. To obtain the forces acting

on our probe for a given flow, we use CFD to find the pressure field by numerically solving the incompressible³ Navier-Stokes equations:

$$\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \vec{u} \quad (7)$$

where $\vec{u}$ is the velocity vector field, p is the pressure, ρ the fluid density, and ν is the kinematic viscosity. Equation 7 is solved via Finite Volume Method (FVM). FVM breaks down the fluid domain into a “mesh” of smaller volumes called “cells”. FVM takes advantage of the divergence theorem, $\int_V \nabla \cdot \vec{F} dV = \int_S \vec{F} \cdot d\vec{S}$, to convert integral quantities into surface fluxes. FVM has advantages over other discretisation methods, like Finite Element Method, because conservation is built-in. As it deals in fluxes across surfaces, it **inherently** satisfies continuity of the transported quantity, ensuring it’s neither created nor destroyed (Figure 5).

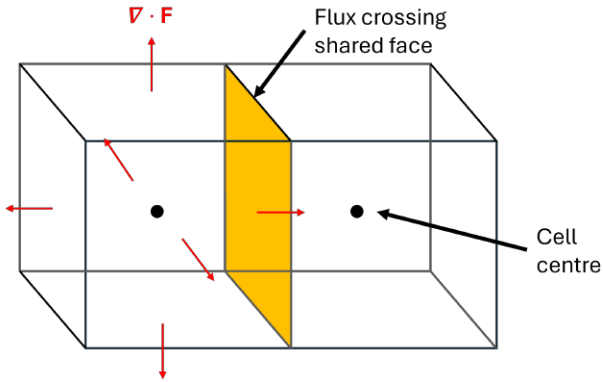


Figure 5: Two neighbouring cells. The face they share is where continuity is directly conserved. Flux leaving one cell through that face **must** equal flux entering the other.

The governing equations are then solved in each cell, and the fluxes of quantities entering and leaving the cell are computed. The values of the quantities are averaged and stored at the cell centre. The

³An advantage of the fact that we are investigating an altitude of Venus at STP is that air density is Earth-like, and at the speeds we expect to encounter we remain in the incompressible regime.

solver iterates on an initial solution until the governing equations are satisfied and quantities are conserved under a tolerance threshold. At this point every cell contains a value of pressure and velocity.

2.3.1 Modelling a Rotating Probe

Our probe is spinning and falling. A full time-evolving simulation of probe motion is too costly, because the timesteps have to be small and the mesh would have to change in each iteration. A static probe, with rotating air moving towards it, is physically equivalent, but avoids these issues. This is the same as positioning ourselves in the **frame of the probe** and observing air approaching and swirling round (Figure 6).

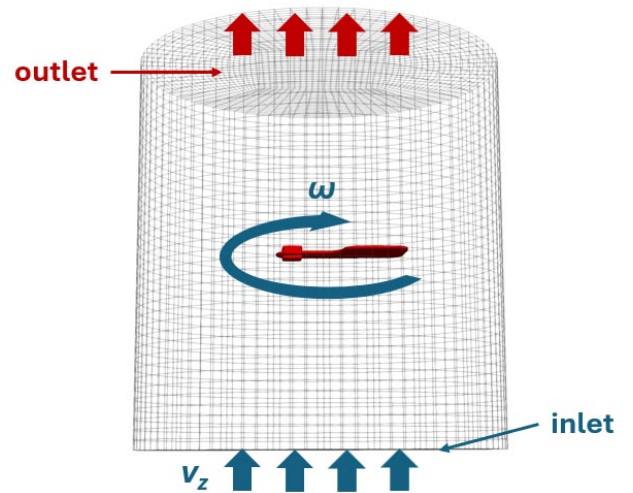


Figure 6: Fixed probe in a fixed cylindrical domain, with air injected at the base, and then immediately rotation is applied (in the direction across the wing from the leading edge). ω and v_z are **input variables** that change with each simulation.

The simulation therefore starts with the probe in rotation, not the at-rest position they will be dropped from. This means we only investigate the steady-state, and that the descent and rotation speeds are *prescribed*. Hence, the simulation is not necessarily a true steady-state. This motivates running **multiple** simulations, and finding the one that most closely

satisfies equations 2 and 3 when combined with the moments due to inertial forces (Section 2.4), thereby identifying the steady-state.

2.3.2 Motivating the Use of CFD

For every cell there is one independent equation. For a system described by one million cells, this leads to a system of one million independent linear equations. Solving this system is computationally expensive, so we must be sure this approach is necessary. There are two reasons why we need CFD. The first is the non-linear $(\vec{u} \cdot \nabla) \vec{u}$ term in the Navier-Stokes equations. This makes analytical solutions impossible in most cases where perturbations to the mean field values are not negligible. If we try to calculate force directly there is another barrier. The equation for drag force is defined as:

$$F_D = \frac{1}{2} C_D A_{rel} \rho v^2 \quad (8)$$

where C_D is the drag coefficient, A_{rel} is area normal to flow, and v is the velocity in the direction against the fluid. C_D is known empirically for some simple geometries, but for our probe that is not the case. This is also a motivation behind running many different simulations, we cannot know beforehand what configuration will lead to $F_D = mg$. These two reasons necessitate the use of CFD, however computationally costly it may be.

2.3.3 Turbulence Modelling

Direct Numerical Simulation (DNS) is the resolution of flow behaviour, including turbulence, at all scales. It has been performed on sycamore seeds in the past [11]. The problem with DNS is that the mesh needs to be incredibly fine in order to resolve the flow at all scales, mesh requirement scales with the Reynolds number to the third power. Sycamore

seed autorotation is reported to have an $Re \approx 100-260$ in the literature, however our probe will operate in a much higher Re regime. Where $U \sim 3 \text{ ms}^{-1}$, $L \sim 0.1 \text{ m}$, and $\nu \sim 1.5 \times 10^{-5} \text{ m}^2\text{s}^{-1}$, the Reynolds number is orders of magnitude greater than the sycamore seeds, at $Re \sim 20000$. As the number of required cell scales with Re^3 , and every cell adds to the N linear equations, this makes DNS unfeasible for our case.

Without DNS we cannot resolve these flows by directly solving eq. 7, instead we can employ a *turbulence model*. Reynolds Averaged Navier-Stokes (RANS) simulations use turbulence models, empirical functions that are capable of evaluating the *average* effects of turbulence without direct resolution. The Spalart-Allmaras turbulence model solves a single additional equation for turbulent kinematic viscosity, ν_t , the additional resistance to flow caused by turbulent eddies, which when combined with kinematic viscosity, ν , can be plugged back in to solve the Navier-Stokes equations [14].

The partial differential equations (PDEs) here are boundary value problems, meaning the solution is constrained at the bounds of the problem, for example, at walls and outlets. The no-slip condition ensures that fluids match the velocity of the surface because of increased viscosity, stopping fluid velocity in one or more directions. This means that the three-dimensional mixing required for turbulence is killed by the boundary, so ν_t goes to zero, and in the far field (away from walls) viscosity is on the order of the ambient molecular viscosity ν_∞ ⁴:

$$\text{At the wall: } \nu_t \rightarrow 0 \quad (9)$$

$$\text{Far field: } 0.21\nu_\infty < \nu_t < 1.29\nu_\infty \quad (10)$$

Near the wall, in the viscous boundary layer,

⁴This is much the same as the standard molecular viscosity ν only it is fixed in the simulation, a physical constant that doesn't vary with temperature.

empirical formulae called *wall functions* are used to evaluate the values as they approach zero. This empirical model is well validated, appropriate for our geometry, and saves computational time.

2.4 Rigid Body Dynamics

The CFD evaluates the forces and moments due to aerodynamics. We must evaluate the moments due to inertial forces⁵ to completely describe the probe motion. Moments of a rotating rigid body are:

$$\vec{M}_{inertia} = \vec{\omega} \times \vec{L} \quad (11)$$

where $\vec{L}$ is the angular momentum:

$$\vec{L} = \mathbf{I}\vec{\omega} \quad (12)$$

In steady-state, we assume the probe will rotate mostly about the z -axis, so that $\vec{\omega} \approx \omega\hat{z}$. Moment of inertia measures a body's resistance to rotation. It is defined as:

$$I_{ij} = \int_V (\delta_{ij}|\vec{r}|^2 - r_i r_j) \rho(\vec{r}) dV \quad (13)$$

where I_{ij} is an element of the inertia tensor, $\vec{r}$ is a position vector, $\rho(\vec{r})$ is the mass distribution, and i, j are indices corresponding to coordinate directions x, y, z . This integral can be evaluated numerically by assigning densities to regions of the probe geometry in Figure 3, and then discretising and summing the mass contributions of every cell. This returns a 3×3 matrix that describes the moment of inertia about the centre of mass. Here we assume the centre of rotation (CoR) is equivalent to the centre of mass, we interrogate the validity of this assumption in Section 3.3.1. This tensor is valid for the probe in the orientation in which it is calculated. So to not

⁵In this work, we refer to “inertial” moments as the combination of moments due to mass distribution and centrifugal forces that represent the fact we are in a non-inertial frame.

have to recalculate the tensor for every set of parameters, we can rotate the tensor to correspond to the rotation of the probe. Rotating the moment of inertia tensor is physically equivalent to rotating an object, then recalculating the new moment of inertia.

$$\mathbf{I}' = \mathbf{R} \mathbf{I} \mathbf{R}^T \quad (14)$$

where $\mathbf{I}'$ is the new tensor, and $\mathbf{R}$ is a rotation matrix combining rotations about all axes (ϕ about x , θ about y). $\mathbf{I}'$ can be used to calculate the moments due to inertial forces in equation 11. We rotate the probe for the CFD simulations about x and then y axes, and this is consistent with the rotation of the tensor.

3 Method

3.1 Parameter Sweep

Steady-state is described by the four parameters in Section 2.1. $\mathbf{I}'$ is calculated about the centre of mass, and our CFD simulations rotate the air about the same axis, so the moments act along equivalent axes and can be summed:

$$\vec{M}_{CFD} + \vec{M}_{inertial} = \vec{M}_{net} \quad (15)$$

We sweep over a number of values of the four parameters⁶, running CFD simulations, evaluating $\mathbf{I}$ and calculating $\vec{M}_{net}$ for every combination. Because of the computational time and numerical uncertainties we sweep parameters from:

1. $-10^\circ \leq \theta \leq 11^\circ$, $\Delta\theta = 1^\circ$
2. $-7^\circ \leq \phi \leq 5^\circ$, $\Delta\phi = 1^\circ$
3. $1 \leq v_z \leq 4$, $\Delta v_z = 0.25 \text{ (ms}^{-1}\text{)}$

⁶We use *rotations per minute* (rpm) and degrees° because it is more physically instructive than radians.

$$4. \quad 120 \leq \omega \leq 300, \quad \Delta\omega = 15 \text{ (rpm)}$$

this produces a dataset of 5832 simulation results⁷. Each data point results from a ‘case’, that is a combination of the four parameters used to run a simulation. These ranges were chosen based on previous students’ work, and our own early experiments. Combining this data with forces and moments evaluated from rigid body dynamics, we can find the combination of parameters that is *closest to* steady-state, where net moments and forces equal zero. The uncertainty in our result would then come from the step size, for example, in cone and pitch angles it will be to the nearest degree.

To compensate for the fact we are only finding the nearest solution for our table of data, we employ **linear interpolation** to search for combinations in-between. Linear interpolation simply draws a line between two points and makes a continuous function out of $y = mx + c$. It is the simplest form of interpolation, and can be less accurate, but more stable than splines. For our case, where the dataset is so dense, the gaps between points are so small they can be treated as linear. The potential for this to be used to reduce the parameter sweep is explored in Section 5.4.

3.2 Numerical Pipeline

3.2.1 Simulation Setup

The list of parameters to run cases with is read from a file. One case is executed per unique combination of parameters. The simulation setup is the same for every case. To run CFD simulations we use OpenFOAM, an open-source toolbox for solving partial differential equations that is widely used

⁷The final dataset is a concatenation of multiple datasets that sweep in slightly different steps, and some areas of parameter space were investigated with far fewer cases, i.e. $\theta < 0$.

for fluid dynamics [15]. OpenFOAM requires three things to run a case:

- Geometry
- Initial and boundary conditions
- Solver scheme.

The design for the geometry was provided by SpaceAM, and turned into a computer model by James O’Sullivan (Figure 3). A geometry alone is not enough, but when combined with a domain and **discretised**, as in Figure 7, this provides the mathematical basis for the solution of the fluid flow with FVM. The discretisation works by taking an existing simple mesh (the polyhedra that make up the cylindrical outer domain), and iteratively “snapping” those cells to the surface of the probe, working outwards to morph the cells making up the cylinder to fit around the probe [16].

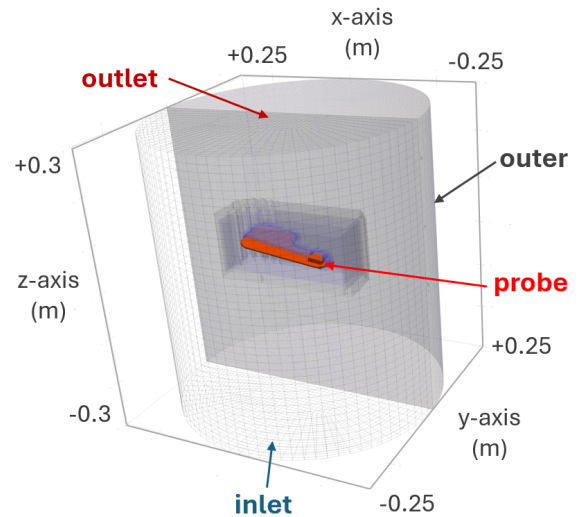


Figure 7: Mesh clipped along the span of the probe (red). Dense collection of cells around the probe is a result of *refinement*, snapping to the surface and increasing the number of cells for improved accuracy. Credit Harvey Reynolds.

Now we have a mesh, but mathematically all the cells are identical. The PDEs here need boundary conditions to be solved. Boundary conditions give surfaces physical meaning, such as the no-slip

Table 1: Boundary conditions on each of the surfaces from Figure 7. SRFVelocity is what causes the flow to appear rotational in the frame of the probe. Pressure here is *kinematic pressure*, that is pressure over density.

Boundary	Pressure (m^2s^{-2})	Velocity (ms^{-1})	ν_t (m^2s^{-1})
Inlet	Zero Gradient	SRFVelocity	Fixed Value
Outlet	Fixed Value	Zero Gradient	Fixed Value
Outer	Zero Gradient	SRFVelocity	Fixed Value
Probe	Zero Gradient	No-Slip	Wall Function

condition which tells the solver this is a solid surface. Zero gradient means the flow doesn't change perpendicular to the boundary. The conditions for quantities discussed so far, on boundaries as labelled in Figure 7, are included in Table 1.

With the boundary conditions applied, we have a closed problem, and a solution to be found by a solver algorithm. The SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm is employed to solve the PDEs in steady-state, where the solution is obtained by *iterating* instead of stepping in time⁸.

3.2.2 Parallelisation

The mesh in Figure 7 consists of over 880000 cells. Running many cases, this would be very time-consuming. We can utilise CPU parallelisation to speed this process up. The domain in the figure can be broken into separate pieces and distributed to all the available processors. The cells on the borders of each piece then communicate with the neighbouring piece to send and receive fluid flux. Our project utilised ISCA, the University of Exeter's resident supercomputer, to maximise computing power by partitioning cases onto different nodes and parallelising each case across 16 processors. Figure 13 visualises the workflow of a single simulation case.

⁸This is why we keep the $\frac{\partial u}{\partial t}$ term in eq. 7, because SIMPLE uses a pseudo-timestep to iterate toward a solution.

3.2.3 Combining Aerodynamics and Mass

The fluid dynamics simulations do not take mass into account. We are searching for the combination of variables that satisfy equations 2 and 3, where mass is a variable. After the CFD simulations return forces and moments, we can evaluate the forces and moments due to mass as per Section 2.4, and combine them. Where the sum of the two is equal to zero, that is our steady-state.

3.3 Drop Tests

Our simulation data is obtained using numerical methods that model real physics. It is necessary to validate the numerical modelling with experiment. To this end, we conducted live drop tests, dropping model probes at height, recorded videos to verify and validate our numerical approach. The test model in Figure 8 was made with a plastic sheet, washers for the 'nut' mass, and wire for the leading edge mass⁹. We conducted many tests, but this report will focus only on probes dropped with *initial rotation* (so they enter steady-state), as this is an investigation into the steady-state flight profile.

Videos of the drop tests were recorded from two different camera angles: one bird's-eye view, and one side-on view (Figure 9). From the top-down view we can extract the centre of rotation, and from

⁹This model was used *only* for centre of rotation analysis, the model in Figure 12 was used to find steady-state parameters.

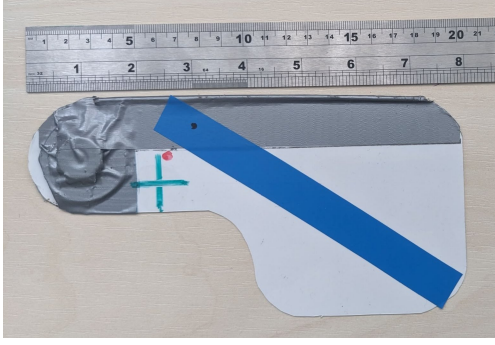


Figure 8: Prototype probe used for the centre of rotation investigation. Mass = 33 grams. Tape length = 15.7 cm.

the side-on view, terminal velocity. From previous students' work we know that we are expecting tens of rotations per second. This makes by-eye analysis challenging. To combat this we employ an eight-times slow-motion camera feature.

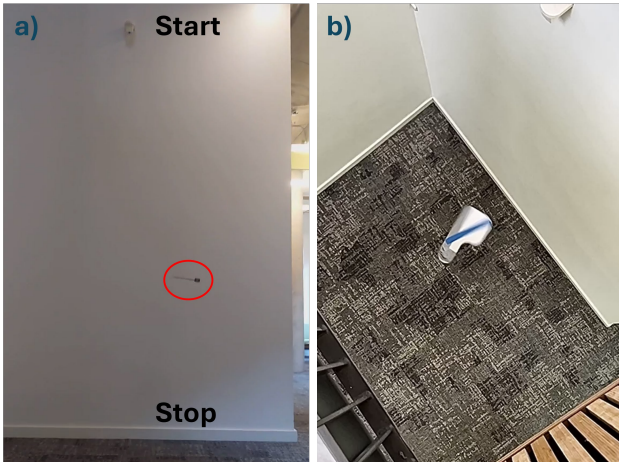


Figure 9: a) view to find v_t . b) Probe recorded from second floor. Dropped from fourth floor (10 m) to settle into steady-state when in frame. Conducted in the Living Systems Institute building at the University of Exeter.

Earlier, we stated an assumption that the centres of mass and rotation lie on the same axis. Equation 15 relies on this being approximately true, however, there is some debate on the validity of this assumption in the literature [6][17]. Aerodynamic forces act away from the CoM, potentially causing an instantaneous centre of rotation away from CoM. We investigate this via drop tests. Even with slow-

motion footage, by-eye analysis was deemed too imprecise and time-consuming for large amounts of data, so we developed computer vision scripts to extract the centre and speed of rotation.

3.3.1 Computer Vision Footage Analysis

Blue tape is applied to the wing in Figure 8 so every frame can track the orientation of the wing. For every video frame, we choose a colour to keep (blue), and turn the rest black, creating an image that is an array of zeroes and ones. Using the location of non-zero indices, algorithms can fit shapes, such as a rectangle, based on angles close to 90° and equal parallel sides. A line drawn through the midpoint of the rectangle then describes the orientation of the probe at a given frame. The rotation speed can be extracted simply from here by taking frames i and $i + 1$, then the angle swept by the probe line between the two frames and the frame rate (fps) gives the rotation speed, $\omega = \Delta\phi \cdot \text{fps}$. The centre of rotation is obtained by taking two frames and mapping lines between two equivalent points. Perpendicular bisectors to these lines then intersect at the centre of rotation, as in Figure 10.

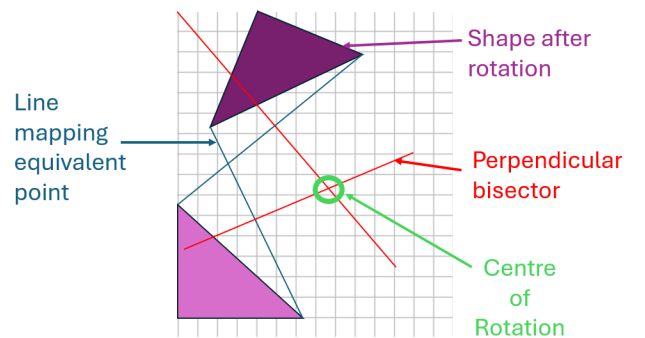


Figure 10: This rotation in space is equivalent to rotation in time. Perpendicular bisectors intersect at CoR.

To compensate for any effects caused by perspective, the centre of rotation is described by a co-ordinate system relative to the tape, parallel distance

along the tape and perpendicular distance from the tape, where distances are relative to the length of the tape in each frame pair. Using many pairs of frames, an average CoR can be determined. This approach was validated on test data to a high degree of accuracy (Appendix A). This analysis automates the analysis of large amounts of footage, and ensures greater accuracy than by-eye. From the side-on view we can obtain v_t by timing falling a fixed distance, and ω , θ and ϕ are obtained from the Inertial Measurement Unit.

3.3.2 Inertial Measurement Unit

SpaceAM provided a model probe with an Inertial Measurement Unit (IMU). An IMU detects its orientation via interaction with Earth's magnetic field (Figure 11). It contains magnetometers that use the Hall effect to measure magnetic field strength. Magnetic field perpendicular to current causes the Lorentz force to accelerate electrons to one side, creating a measurable potential difference. The strength of this voltage relates to how aligned the magnetic fields of the IMU and Earth are. Oscillation of directions in x and y can be used to determine rotation speed.

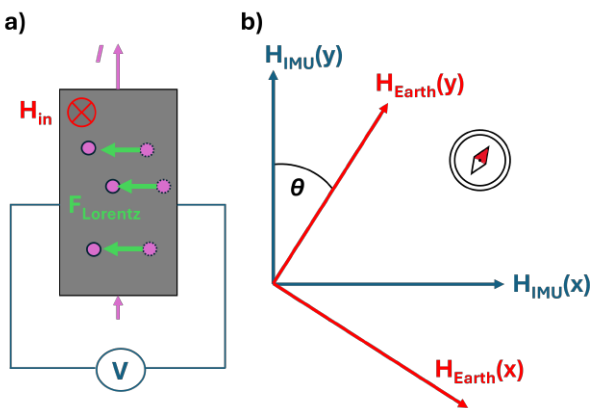


Figure 11: a) Single magnetometer showing pink electrons shifting due to Lorentz force. b) Relative orientation of the IMU to Earth's magnetic north.

This device supersedes video analysis for deter-

mining angles and rotation speed, but we still require videos for terminal velocity and centre of rotation analysis. Figure 12 shows the model provided by SpaceAM with the IMU where the “box” was in Figure 3. Note that CFD simulations do not model the intricacies of the hardware (such as screws) as this would lead to a mesh of many millions of cells, so the box acts an approximation. The pipeline for the experimental evaluation of steady-state is presented in Figure 15.



Figure 12: SpaceAM's prototype probe design with IMU unit. Mass = 28.1 g.

Even with the experimental hardware, a working numerical pipeline is still valuable because it can help to rapidly iterate over different designs, whereas producing different models with different IMUs and then calibrating them takes significant time investment.

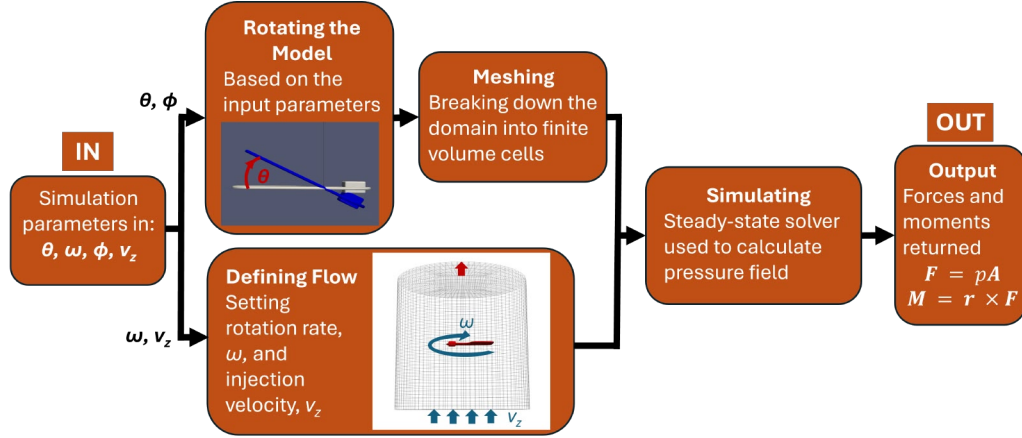


Figure 13: Workflow of one simulation case. The combination of input variables is unique to each case. Outputs forces and moments in x, y, z from aerodynamics.

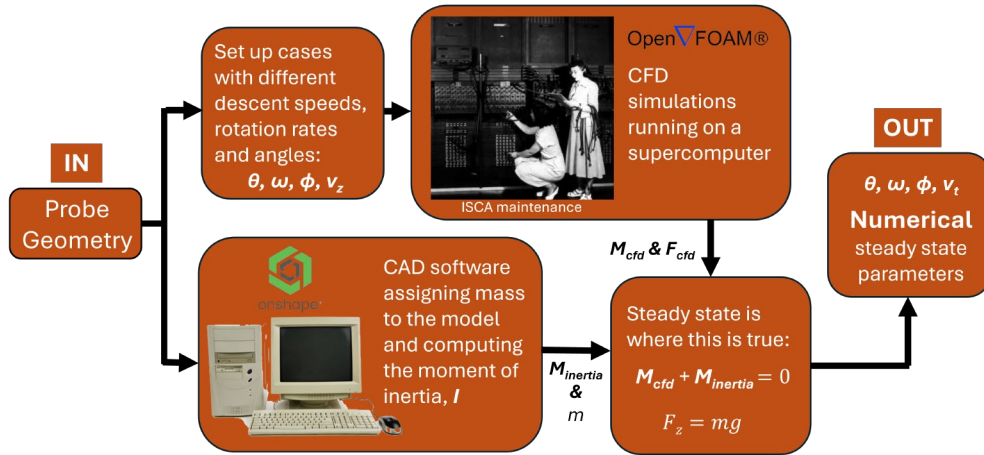


Figure 14: Numerical workflow that takes a given probe design and returns the steady-state parameters. Simulations run as per Figure 13. Onshape is the software that calculates I .

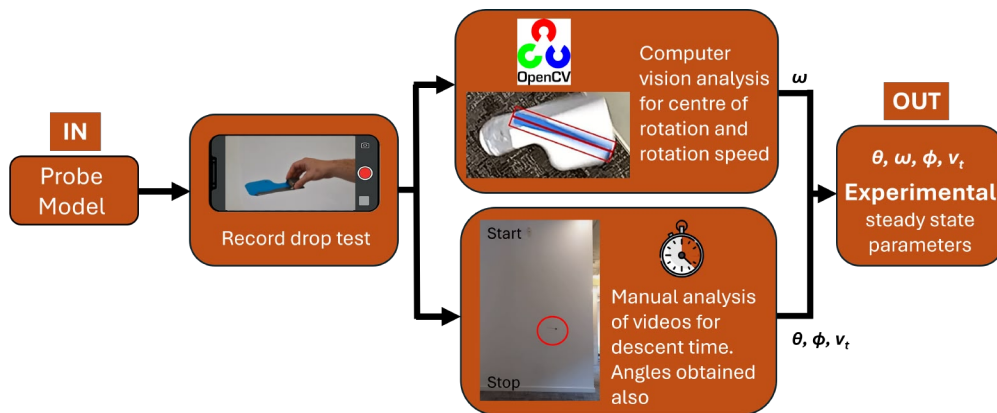


Figure 15: Pipeline for conducting drop tests to extract steady-state variables. OpenCV is the computer vision library used in video analysis. The results in this work report angles and rotation speeds as obtained by IMU (Section 3.3.2).

4 Results

4.1 Steady-State Flight Solution

Each simulation calculates forces and moments acting on the probe. Forces are evaluated from the pressure field as calculated from eq. 7 at each iteration (Figure 16). The SIMPLE algorithm uses a pseudo-timestep to converge on a steady solution, so initial field evaluations are often not realistic. For this reason, we take an average of the values once they settle (Table 2). Figure 17a shows the values of the forces and moments for one specific case, over the course of the simulation.

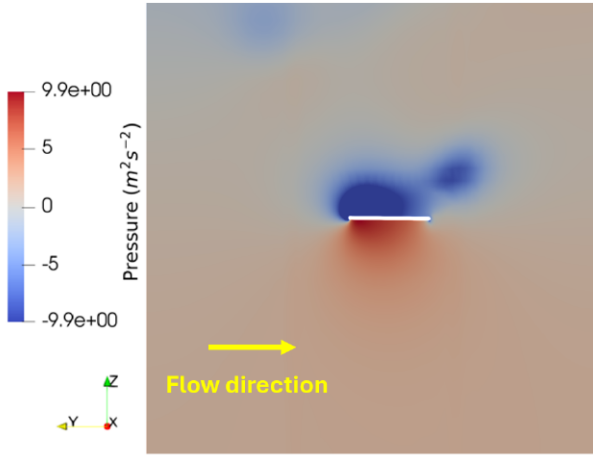


Figure 16: Pressure distribution on a chord-wise slice of the domain. Low pressure region above wing increases lift.

A simulation will stop iterating once it reaches the specified convergence tolerance (previous result similar enough to the current one). With a maximum 2000 iterations, very few simulations met this criteria. This is because the solver converges around a value and then oscillates, residuals increasing and then decreasing about a value but never fully converging to the point of satisfying the specified tolerance, as in Figure 17b. The implications of this behaviour are discussed in Section 5.3.

An example output can be seen in Table 2, the moments here would then be combined with mo-

ments as calculated by eq. 11, $\vec{M}_{inertial}$, which are a function of the parameters: rotation speed, cone and pitch angles. Forces in x and y are not given here, because for an object rotating purely about the z -axis the sum of forces in x and y sum to zero over one rotation.

Every simulation results in a row of forces and moments due to aerodynamics in the dataset (Table 2). Combining the dataset provided by CFD simulations with inertial forces allows us to identify the steady-state, from all the combinations of the four parameters. Table 3 provides the steady-state parameters for our test probe, as calculated from our numerical pipeline, and from drop tests.

The results from the numerical pipeline represent the state that is *closest* to steady from our table of values. There is still a net moment and force in this case, and those values are shown in Table 5. True steady-state satisfies the condition that net moments and forces equal zero. Using interpolation, we investigate the possibility of getting closer to a true steady-state without running more simulations. Table 4 shows the results of interpolation compared to values straight from the data table, and net forces and moments in Table 5 show how improved the interpolated result is. The reliability and predictive power of the interpolated model is explored in Section 5.4.

4.2 Varying Probe Mass

The primary objective of this project was to create, and then verify, a numerical pipeline to accurately characterise steady-state for a given probe design. We demonstrate this capability in Table 3. However, we look to see if we can effectively use this methodology for more probe designs. In keeping the geometry the same and only varying mass distribution

Table 2: One case of CFD output. Input parameters on the left, output results on the right. M_i is the moment about the i -axis. Moment units are mN·m (milli-Newton metre) and force units are mN. Resulting from *only* aerodynamics.

θ (°)	ϕ (°)	ω (rpm)	v_t (ms ⁻¹)	M_x	M_y	M_z	F_z
10	0	400	3.00	-2.0	-17.1	0.7	259.0

Table 3: Steady-state parameters determined from both the numerical pipeline and experimental drop tests.

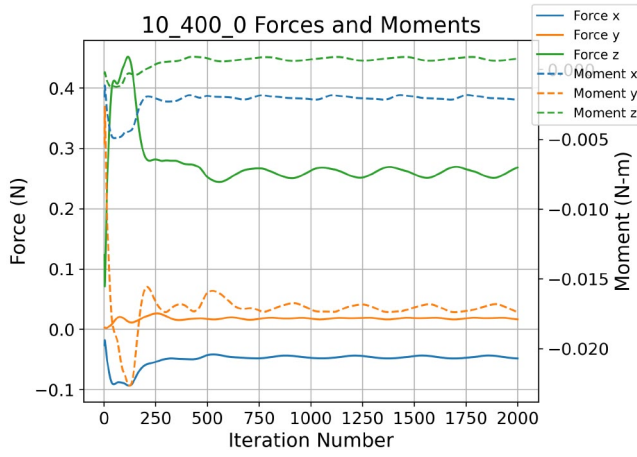
Method	θ (°)	ϕ (°)	ω (rpm)	v_t (ms ⁻¹)
Simulation	5 ± 1	-4 ± 1	450 ± 15	3.25 ± 0.25
Experiment	9 ± 3	-2 ± 1	412 ± 23	2.9 ± 0.3

Table 4: Steady-state parameters determined from the data table and from linear interpolation performed on that same table.

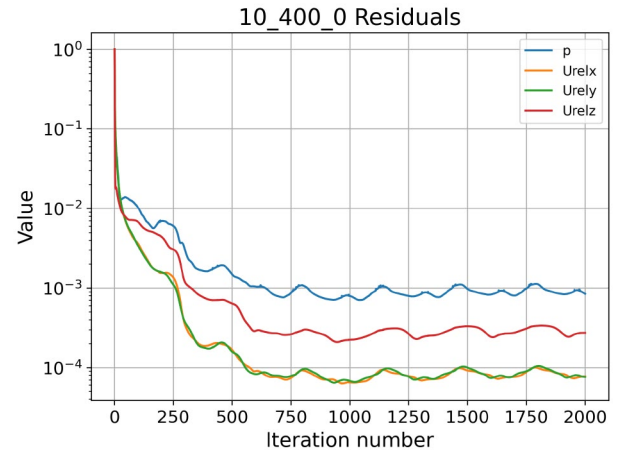
Source	θ (°)	ϕ (°)	ω (rpm)	v_t (ms ⁻¹)
Data Table	5 ± 1	-4 ± 1	450 ± 15	3.25 ± 0.25
Interpolation	5.2	-3.1	449	3.18

Table 5: Net moments and forces for the test probe in closest-to steady-state, as found from the data alone, and with linear interpolation. μN is micro-Newton.

Source	M_x ($\mu\text{N}\cdot\text{m}$)	M_y ($\mu\text{N}\cdot\text{m}$)	M_z ($\mu\text{N}\cdot\text{m}$)	F_z (μN)
Data Table	167	-553	1727	1208
Interpolation	-8	15	1670	-10



(a) Forces and moments evaluated at each simulation iteration.



(b) Residual difference from prior iteration, where p is the pressure and Urel is the relative velocity in x, y, z components.

Figure 17: Values and their residuals at each iteration for $\theta = 10^\circ$, $\phi = 0^\circ$, $\omega = 400$ (rpm) and $v_z = 3 \text{ ms}^{-1}$.

we can use the same table of aerodynamics results¹⁰. The new moment of inertia is calculated by eq. 13 and the new mass is an input.

Designs with additional mass on the box, and mass along the wing, were tested. Terminal velocities of all three designs, dropped with either the largest “box” volume above the wing (up) or below (down), are in Table 6. Orientation data was unavailable for the wing added mass model, but a comparison of standard probe and added box mass is present in Table 7, as well as accompanying numerical predictions for the added mass probe.

4.3 Centre of Rotation Analysis

The validity of previous section’s results rely on the centres of rotation and mass aligning. We conducted video analysis aided with computer vision techniques to determine the CoR of a test probe. We apply the method in Figure 10 to slow-motion footage of drop tests as in Figure 18, where the centre of rotation is identified by the orange circle, which is at the intersection of the perpendicular bi-

¹⁰This does mean that the centre of rotation has changed, so with the same CFD data we can only faithfully model new probes with uniform mass change.

Table 6: Terminal velocity for different models. Control is the probe in Figure 12, “Nut+” has mass added to the IMU, and “Wing+” has added mass at the IMU **and** the mass of the wing was doubled. Model mass, m , in grams.

Model	Orientation	Terminal Velocity (ms^{-1})
Control	Up	2.9 ± 0.3
$m = 28.1$	Down	2.8 ± 0.3
Nut+	Up	3.4 ± 0.3
$m = 38.3$	Down	2.9 ± 0.3
Wing+	Up	4.8 ± 0.4
$m = 46.8$	Down	4.2 ± 0.4

sectors in light-blue. The position of the probe in the previous frame is the darker red line, and the lines that map the translation of equivalent points are green. That darker red line can then be used to determine the rotation speed as in Figure 23 which provides validation for our numerical pipeline results.

The result in Figure 19 shows that, with experimental errors, the assumption that centres of rotation and mass are the same is valid, supporting the methodology that we have used to derive values in Tables 2-7. This degree of accuracy is suit-

Table 7: Steady-state parameters determined for the probe of increased mass. Numerical values from interpolation. Model 1 values as reported in Table 3. Model 2 has added mass on IMU. Model mass, m , in grams.

Model	Source	θ ($^\circ$)	ϕ ($^\circ$)	ω (rpm)	v_t (ms^{-1})
Model 1	Simulation	5.2	-3.1	449	3.18
$m = 28.1$	Experiment	9 ± 3	-2 ± 1	412 ± 23	2.9 ± 0.3
Model 2	Simulation	4.4	-1.6	542	3.58
$m = 38.3$	Experiment	11 ± 4	-3 ± 3	435 ± 37	3.4 ± 0.3

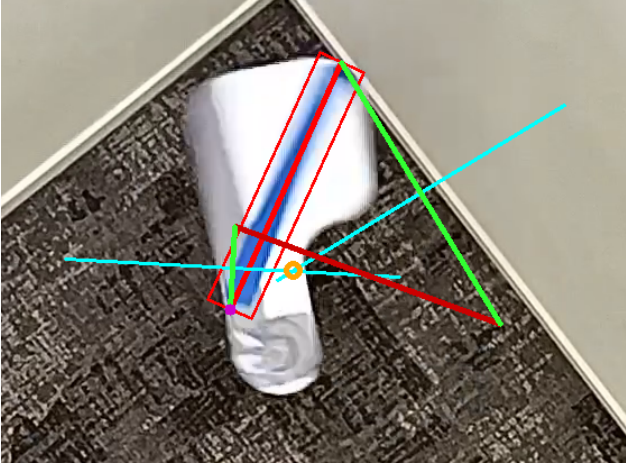


Figure 18: Still capture of the video output that tracks the CoR during descent. Orange circle identifies the CoR between two frames at this instant.

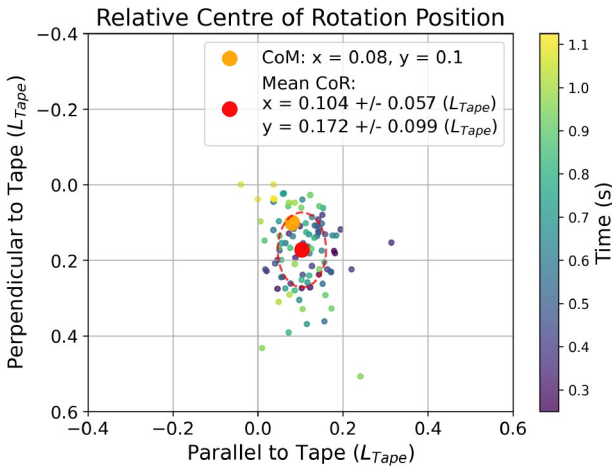


Figure 19: Averaged rotation centre identified in red. Other points correspond to rotation centres identified over the course of the video. Centre of mass in orange, uncertainty is dashed red ellipse. $L_{Tape} = 15.7$ cm.

speed camera, this could be more conclusive. The result here demonstrates the capability to evaluate the centre of rotation and rotation speed effectively.

5 Discussion

5.1 Validation

We validate our numerical result with experiment. However, it should also be possible to validate the data based on expected physical trends. One such trend is the relationship between drag force in the z -direction and descent velocity. We expect the square of velocity, v_z^2 , to relate linearly to F_D as per equation 8. Physically, you would expect the faster air to produce more force on the object. Analysing the data we see our numerical results are physically consistent and the linear relationships are strong, with R^2 values greater than 0.95 in Figure 20¹¹.

The simple model in equation 4 can further help us validate our data. According to the model, for a fixed cone angle, net M_y should increase linearly with the square of ω . Figure 21 demonstrates that this relationship holds to a good standard. Deviations from the linear relationship as v_z increases can be explained by the second v_z^2 term in equation 4. Physically, this represents aerodynamic forces hav-

able for our proof-of-concept, however, with minor improvements, such as bright lighting and a higher

¹¹ R^2 is the coefficient of determination and is a measurement of goodness of fit, in this case how linear it is, with 1.0 being perfect and <0.7 being poor.

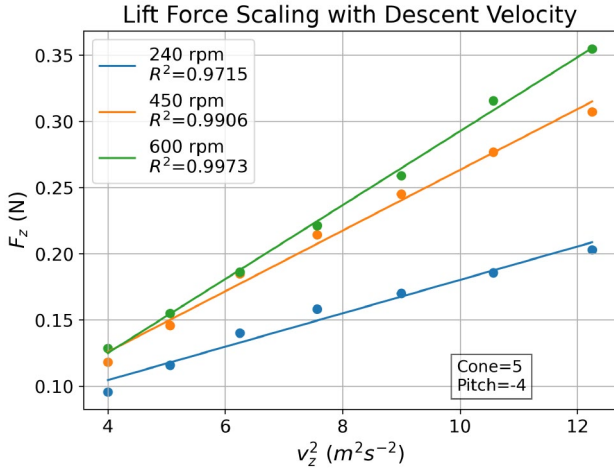


Figure 20: Validation test for forces and velocity in the z -direction. Points represent data and lines represent linear fits, each for different ω values.

ing increased impact on the net result, which for low v_z is dominated by inertial forces. This is also consistent with our understanding that increasing ω decreases θ , acting to flatten the probe's flight profile, demonstrated here by increasing positive M_y .

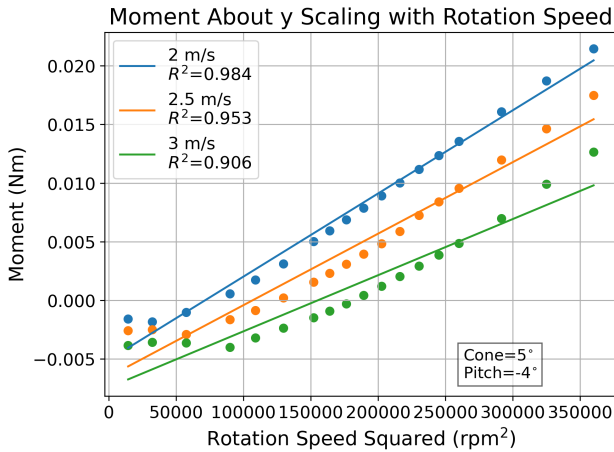


Figure 21: Validation test for increasing M_y with the square of ω for different v_z values. Points represent data, lines are linear fits, each for different v_z .

Additional validation is achieved with analysing individual CFD simulations. They were internally validated by checking the pressure distribution is as predicted by the literature (Figure 16), rotation of flow is consistent, and the velocity incident underneath the probe is equivalent to the descent velocity,

so we are comparing like-for-like (Appendix B). These validation checks give us confidence that the simulations are correctly modelling the aerodynamics, and that we are correctly accounting for inertial forces when calculating net moments.

5.2 Characterising the Steady-State

Table 3 reports the results of the minimised forces and moments, the closest of our simulations to steady-state. To find this solution, we developed a code that uses a least-squares algorithm to find the minimal variable combination. Pseudocode illustrating how that works, and links to the code are in Appendix C. Acceleration due to gravity is a parameter here, so by adjusting this and the fluid viscosity in CFD simulations, this work is easily transferable to Venusian conditions.

Least-squares is used in place of a root finder like Newton-Raphson because there exists no crossing with zero net forces and moments for any combination of parameters. Figure 22 is a slice of the data at the point in parameter space that most closely satisfies a solution. At no point do all four lines cross zero at the same place, and M_z does not cross at all. The implications of this are discussed in Section 5.3.

The results for steady-state parameters from the numerical pipeline in Table 3 agree with the drop tests within experimental and numerical uncertainties for all quantities, bar one. Our findings of positive cone and negative pitch angles are consistent with the literature on autorotating fliers [18]. This result validates the approach taken in this work, and the ability of the produced numerical pipeline to predict steady-state flight parameters.

Deviations from experiment may be explained by shortcomings in experimental validation, not the numerical work. Figure 19 shows rotation speed

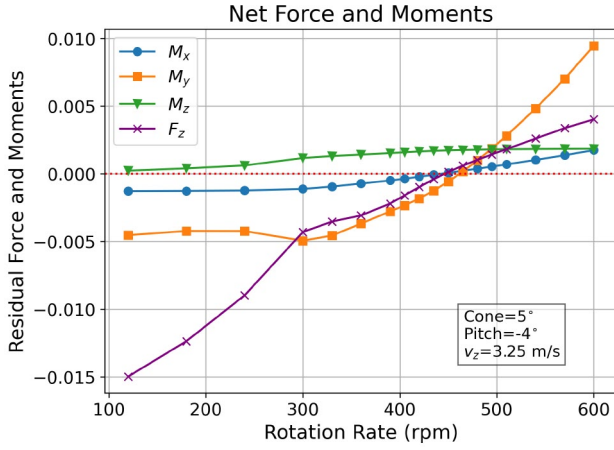


Figure 22: Net force and moments acting on the probe for a slice of the data around our solution, $\theta = 5^\circ$, $\phi = -4^\circ$, $v_z = 3.25 \text{ ms}^{-1}$.

for the prototype probe used in the centre of rotation analysis over time (Figure 8). This probe was dropped from 10 metres with initial rotation. Rotation speed is evidently still increasing before it hits the floor (the frames after the end of the graph).

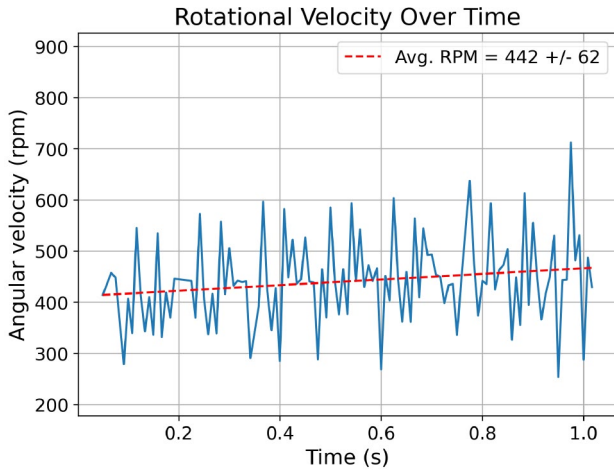


Figure 23: Rotation speed of probe in Figure 8, with a linear fit to the data (red). Gradient is 55 rpm/s. Legend reports mean and standard deviation.

The rate of increase in rotation speed in the second before the probe hits the floor is 55 rpm/s, suggesting that given more time in flight, the probe might have tended towards the state described by our results. The relationship in equation 6 shows that increasing ω decreases θ , which would account

for the cone angle value found by the numerical workflow being outside of the experimental uncertainties. If steady-state was **not** reached in the drop tests, this would explain a systematic overestimation by our numerical model.

5.3 Evidence of an Unsteady Solution

The primary objective of this work is to identify a steady-state flight configuration, defined by simultaneous satisfaction of equations 2 and 3. However, analysis of the dataset reveals that the closest possible candidates for settled flight configurations have a significant residual moment about the z -axis (Table 5). This has implications for the existence of a steady-state for this system.

Non-zero moments about the z -axis are akin to net torques driving angular acceleration. Table 5 shows that for our minimised solution, the net force is negative, gravity dominating lift. A system may drive a greater rotation speed, $M_z > 0$, in order to produce more lift. This is because rotation speed and lift have a positive relationship, shown in Figure 24 for different descent speeds.

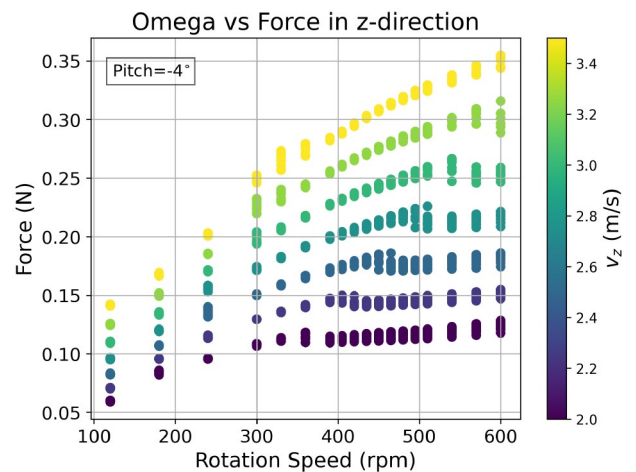


Figure 24: Increasing lift force with rotation speed. Colour represents descent velocity.

From the literature and our own observations we see that autorotating fliers prefer a negative pitch

angle (leading edge pointing down into the flow). In our coordinate system, positive moments about x cause the pitch angle to become more positive. Figure 25 shows that for a given pitch, the faster rotation rates produce *more* of a positive moment about x , levelling out the pitch angle. We also see a turning point about $\phi = 0^\circ$, after which faster rotation acts to put a more *negative* moment about x , to restore the system to a negative pitch angle.

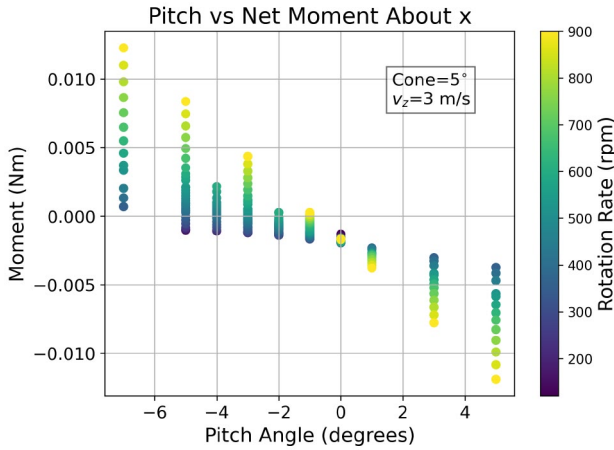


Figure 25: Pitch angle against moment in the x -direction. Colour represents magnitude of rotation speed.

The relationship in Figure 26 shows the use of pitch as a “braking” mechanism. Positive pitch angles demonstrate negative M_z , acting to decelerate the angular rotation. If faster rotation rates act to increase pitch angle, and increasing pitch angle decelerates rotation, this suggests the possibility of a *quasi*-steady state, where restoring forces keep the flight settled but constantly oscillating. Any system that is accelerating cannot be said to be in perfect steady-state. This leads us to reframe our aim, to finding the *settled* flight profile.

The idea that this is a system of changing values is supported by the results of the individual CFD simulations. They display oscillating, “saw-tooth”, residual behaviour, as in Figure 17b, indicative of a truly transient system being modelled with a steady-state solver [19]. The solution stops converging be-

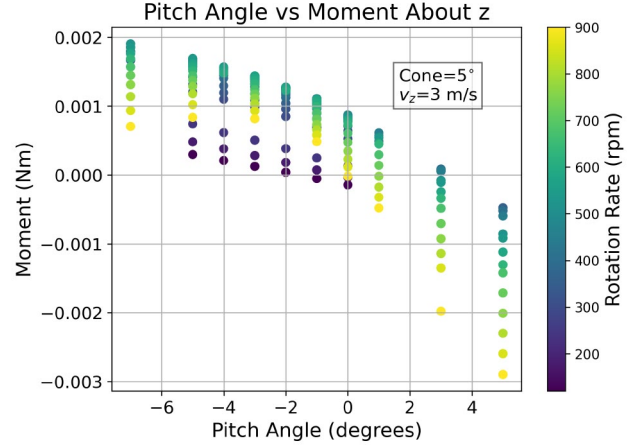


Figure 26: Pitch angle against moment in the z -direction. Colour represents magnitude of rotation speed.

cause it is impossible to pin down a correct solution when the system is, in fact, constantly changing.

However, a true steady-state might still be possible. The inertial forces that combine with the aerodynamics to make the net forces and moments are calculated based on the assumption that the system rotates exclusively about the z -axis. This makes the contribution to the moment about z from inertial forces equal to zero, as per equation 11. If this were not the case, then an equilibrium position where $\sum M_z = 0$ might be possible. This would be in agreement with recent literature on samaras that states that the angles of orientation are neither constant nor small [20]. If the *rates of change* of pitch and cone angles are constant, then a steady-state might still be found.

5.4 Towards a Surrogate Model

The CFD simulations that provide the aerodynamic forces and moments take a long time to run, and feasibility depends on available computational resources. The industry standard approach is to use available data to look for trends to train a surrogate model on, that can give solutions without the attached computational cost [21]. This model can

come in the form of machine learning, but for our project we investigate the use of interpolation. This is not the same as training a model, but can demonstrate *predictive capabilities*.

The dataset obtained from the parameter sweep in Section 3.1 covers a wide range of variables at fine steps. We want to find out how sparse the dataset can be and still return a valid solution. Linear interpolation is used here, because of the multi-dimensional nature of the data and relatively linear relationships between data points, it was found to be superior to other methods, such as splines. We conduct tests to remove data points and see how this affects results. Figure 27 shows the mean difference in moments between interpolated function and data points for a slice of the data, as the percentage of available data is reduced. The interpolated functions are constructed out of the surviving data only via SciPy's `LinearNDInterpolator`.

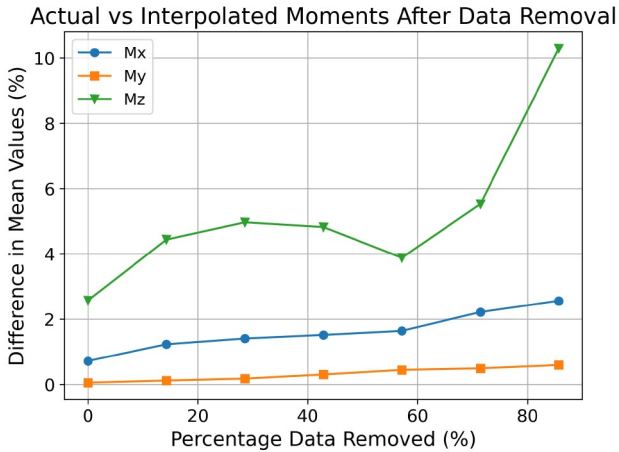


Figure 27: Percentage mean difference between moments in x, y, z at data points and the interpolated function for $\theta = 3^\circ$, $\phi = -1^\circ$, $v_z = 3 \text{ ms}^{-1}$, as data provided for interpolation is reduced.

Results show that, even with substantial data reduction, the global error in moment values remains relatively small. However, small local error can lead to significant differences in the predicted steady-state configuration (Figure 28). This is because the steady-state is determined by locating a minimum

in net force and moment, making it sensitive to perturbations in the data around said minimum. However, it demonstrates that reduced spaced sampling can still return the same answer with interpolation. However, it is best used as a refinement tool, rather than an outright replacement of large subsets of parameter space.

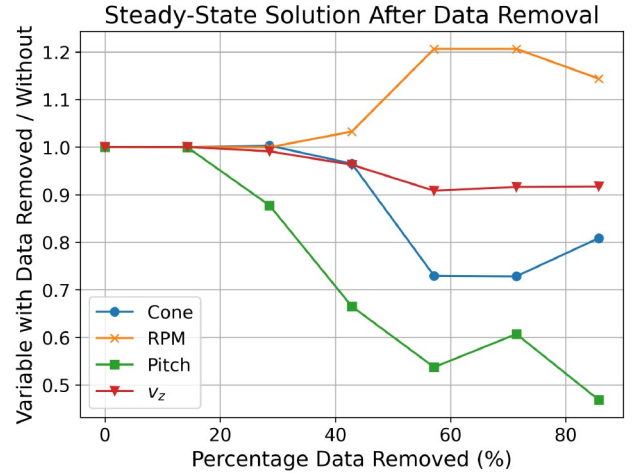


Figure 28: Steady-state variables calculated over increasingly sparse datasets. Variables as a ratio of the calculated value (sparse data) over the value found from the entire dataset.

5.5 Future Work

The probe in Figure 3 is antisymmetric about the plane of the wing. The box containing the hardware sticks out more on one side than the other. According to the inertial forces and CFD, dropping the probe on either side should produce nearly identical results, although all data here is reported for more volume facing “up”. Table 5 shows that for drops with initial orientation “down” terminal velocity is lower. This is likely explained by increased mass about the IMU reducing cone angle, increasing rotation rate, and hence lift, but a study into varied mass distributions would be more conclusive.

Additional modifications to the mass of the probe were not possible because they then did not

reach near steady-state flight in the 10 m drop, and Figure 23 is evidence of this. Drop tests should be performed at greater heights to properly validate the numerical pipeline, and allow for more massive designs to be tested.

We identify that the settled flight is inherently unsteady, therefore transient, or even full 6 Degrees-of-Freedom simulations, such as in Lolli [22], would be a step towards better capturing the physics here. These simulations would have to be “initialised” from a steady-state solution.

6 Conclusion

This project developed a methodology to numerically evaluate the steady-state flight parameters of an autorotating probe design that will be used to explore and sample the atmosphere of Venus. The methodology utilises CFD and rigid body dynamics to construct a large look-up table of forces and moments that can be used to determine the steady-state for a given probe design. The results of this method for a test probe are given in Table 3. These results were validated by carrying out experimental drop tests, and by drawing on our understanding of simple physical models (Section 5.1).

Drop tests confirmed a probe with a large 10 cm chord length is capable of autorotation, even at a higher Reynolds number than other models in the literature [12][23]. Additionally, they confirmed our assumption that the centres of mass and of rotation align, within experimental uncertainties (Figure 19). However, it is unlikely that any of the probes reached settled flight, even in the 10 metre drop. Figure 23 shows that the rotation speed was still increasing. This makes drop tests from greater heights a target for future work, as well as proving the usefulness of a validated numerical pipeline, where different models can be tested *without* needing the nec-

essary lab space.

Interrogation of the data suggests that a “steady-state” as defined in equations 2 and 3 may not ever truly exist, contradicting earlier literature, but supporting similar recent findings for autorotating winged seeds, only in our case for higher Reynolds number flow [20][24]. We propose that the system maintains a quasi-steady flight, where the variables of rotation speed and orientation angles are constantly oscillating, kept in settled flight by restoring forces (Section 5.3). The potential of interpolation to reduce the parameter sweep and build towards a surrogate model was explored, Figure 27 shows that large amounts of the original dataset can be removed with < 10 % trade-off in accuracy. This work is successful as a proof-of-concept for a numerical pipeline that integrates CFD simulations with inertial forces, to numerically determine the steady-state flight profile of a given probe design for use in a mission to explore Venus.

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Appendices

Appendix A - Validation of Video Analysis

Video analysis of centre and speed of rotation was validated using synthetic data, created with the OpenCV library. A video of a blue line rotating about an also moving centre of rotation is created to be analysed.

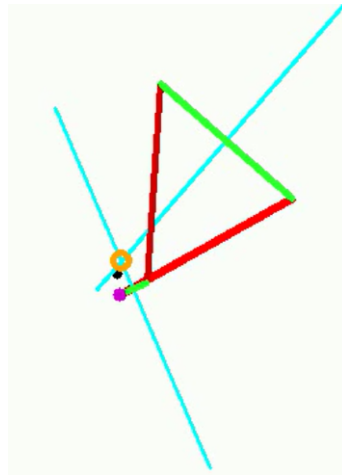
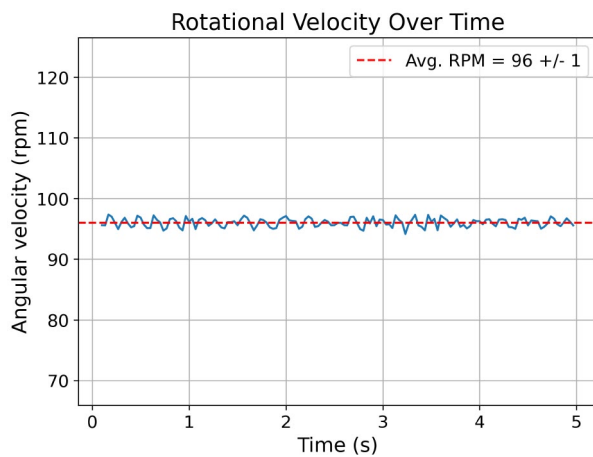
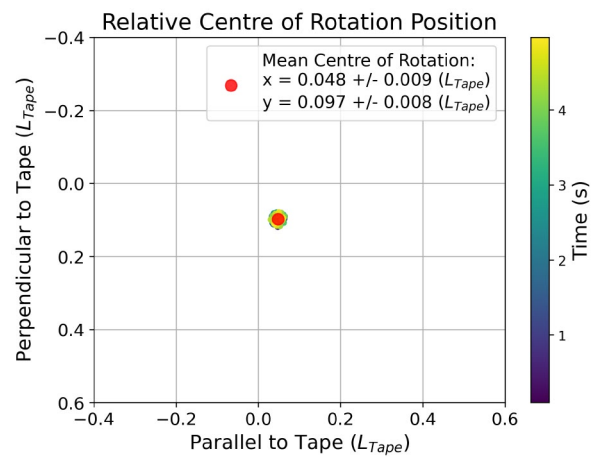


Figure 29: The rotating line appears red here because of the tracking line running down the length of the “rectangle” (blue line).



(a) Rotation speed for synthetic data.



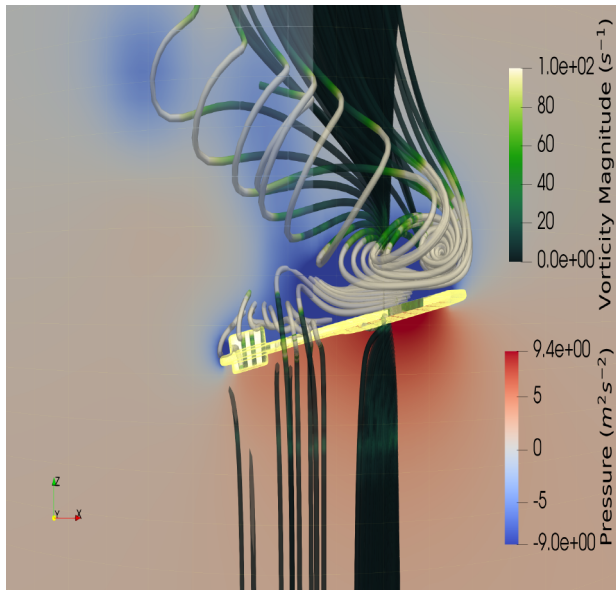
(b) Centre of rotation analysis. Converges on real value within uncertainty.

Figure 30: Results for synthetic data video analysis.

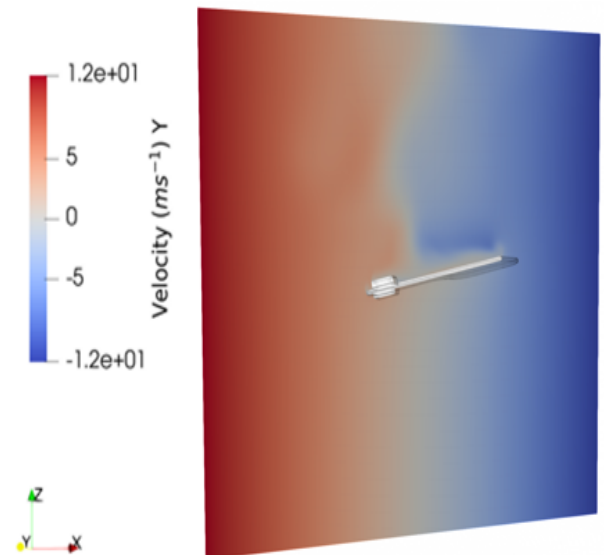
The figures show good agreement with the true centre of rotation, and very good agreement with the true rotation speed (test data uses 8 rotations in 5 seconds = 96 rpm), giving us confidence in using this method on real test data.

Appendix B - CFD Case Validation

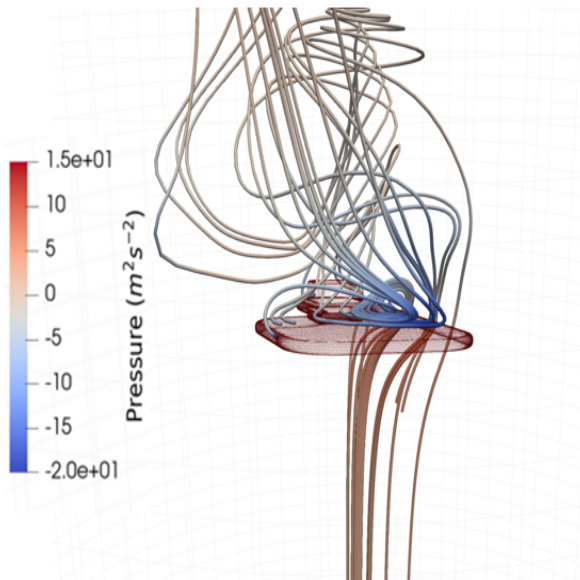
Individual CFD simulations were validated by analysing the output fields at the final iteration for physically consistent behaviour (Figure 31).



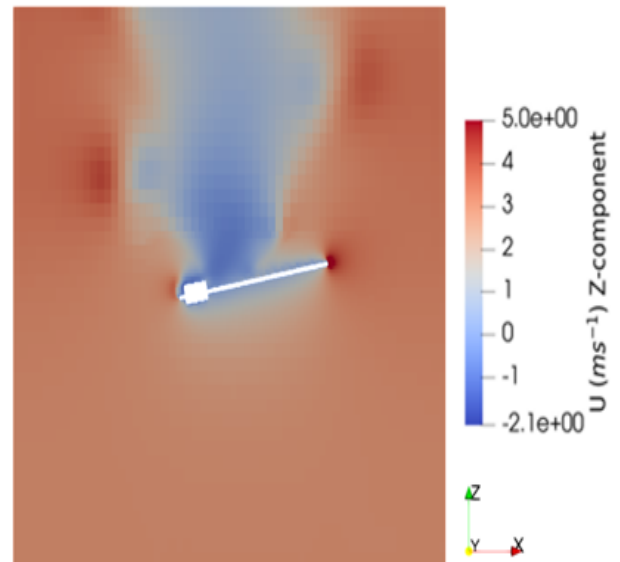
(a) Vorticity against spanwise pressure distribution slice backdrop. Vorticity is defined as the curl of the fluid velocity, and high vorticity is a condition for vortex formation.



(b) y-component of velocity ensures that the flow is correctly rotating around the cylinder, -ve in Y, over the wing from leading to trailing edge.



(c) Streamlines represent paths that a parcel of fluid will take when placed in the flow. Coloured by pressure at those points. Some low pressure vortex is partially visible above the wing.



(d) z-component of velocity, spanwise slice.

Figure 31: CFD outputs for the 10 degree cone, 400 rpm, 0 degree pitch, 3 m/s descent case.

Appendix C - The Code And The Data

Below is pseudocode that illustrates the workings of the code `minimiser.py`:

```
! weight of the probe
mass = 0.029
g = 9.81
weight = -mass * g

! CFD forces and moments stored in csv file
cases = load_data(dataset)

difference = []

for case in cases:
    cone, omega, pitch, vz = case

    ! rotates moment of inertia and does  $M = \omega \text{ cross } I \cdot \omega$ 
    M_inertia = func_inertia(cone, omega, pitch)

    ! get moments and forces due to CFD
    M_CFD, Fz = lookup_table(case)

    M_net = M_inertia + M_CFD
    F_net = (Fz + weight) * scaling_factor

    difference.append(sqrt(M_net**2 + F_net**2))

! finds the index of minimum difference
min_idx = argmin(difference)

solution = cases[min_idx]
```

It is important for future work to mention the scaling factor. The magnitude of the forces is usually an order of magnitude greater than the moments. This means that in seeking a least-squares solution, forces will outweigh moments in “preference”. To combat this, a scaling factor was introduced. This factor is a numerical trick, and was **not** used in the results given here. It was not used because it led to a much greater overestimation by numerical results in comparison to experiment. However, if Section 5.2 is correct, and the experiments did not reach steady-state, then this numerical trick might actually be providing a more valid solution.

To find code and datasets relating to the numerical pipeline probe analysis refer to the GitHub repository below:

<https://github.com/Riaan-jansen/Venus-Life-Finder-Probe.git>

For the video analysis code, including test footage and example results:

<https://github.com/Riaan-jansen/Probe-Video-Processing.git>

Both have READMEs that should be able to point you in the right direction.